B.Tech CSE - Third Year | Lab 5

Assignment 5: Linear Search • *Aim: Search values sequentially and count occurrences.*

Experiment No. 5.1: Linear search when key is present

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_01_Linear_search_when_key_is_present {
    static int linearSearch(int[] a, int key) {
        for (int i = 0; i < a.length; i++) {
            if (a[i] == key) return i;
        }
        return -1;
    }
    public static void main(String[] args) {
        int[] a = {45, 67, 80, 52, 90};
        System.out.println(linearSearch(a, 80));
    }
}
```

3. **Output:** `2`
4. **Viva-Voce:** What does linear search return when a key is found? Why return -1? Must the array be sorted?

Experiment No. 5.2: Linear search when key is absent

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_02_Linear_search_when_key_is_absent {
    static int linearSearch(int[] a, int key) {
        for (int i = 0; i < a.length; i++) {
            if (a[i] == key) return i;
        }
        return -1;
    }
    public static void main(String[] args) {
        int[] a = {45, 67, 80, 52, 90};
        System.out.println(linearSearch(a, 100));
    }
}
```

3. **Output:** `-1`
4. **Viva-Voce:** What does -1 indicate? What is the worst case for linear search? Can linear search handle unsorted arrays?

Experiment No. 5.3: Count occurrences of 52

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_03_Count_occurrences_of_52 {
    public static void main(String[] args) {
        int[] a = {52, 10, 52, 7, 52};
        int count = 0;
        for (int value : a) {
            if (value == 52) count++;
        }
        System.out.println("Count: " + count);
    }
}
```

3. **Output:** `Count: 3`
4. **Viva-Voce:** What is an occurrence? Why not stop at the first match? Which condition checks equality?